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Exercise-Based Prevention of Musculoskeletal Disorders in Dental Professionals

Aleksandra Dobrzańska, ORCID <https://orcid.org/0009-0002-4694-793X>

aleksandra.dobrzanska89@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,

ul. Pomorska 251, 92-213 Łódź, Poland

Aleksandra Baranowska, ORCID <https://orcid.org/0009-0003-0373-7577>

olabaranowska200@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,

ul. Pomorska 251, 92-213 Łódź, Poland

Klaudia Borycka, ORCID <https://orcid.org/0009-0006-0006-7813>

k.borycka@interia.pl

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Maria Berek, ORCID <https://orcid.org/0009-0006-2613-1123>

mariaberek12@gmail.com

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Anna Hajduk, ORCID <https://orcid.org/0009-0003-1570-0608>

hajduk.ania2001@gmail.com

Independent Researcher, Łódź, Poland

Kacper Kmieciak, ORCID <https://orcid.org/0009-0007-2703-2538>

kacper.kmieciak@o2.pl

Institute of Dentistry of the Central Clinical Hospital of the Medical University of Lodz,
ul. Pomorska 251, 92-213 Łódź, Poland

Filip Kostanek, ORCID <https://orcid.org/0009-0003-0243-5219>

filipkostanek@gmail.com

Independent Researcher, Łódź, Poland

Jagoda Stańska, ORCID <https://orcid.org/0009-0000-4861-2856>

lek.dent.jagoda.stanska@gmail.com

Institute of Dentistry of the Central Clinical Hospital of Medical University of Lodz,

ul. Pomorska 251, 92-213 Łódź, Poland

Piotr Szymański, ORCID <https://orcid.org/0009-0006-2891-640X>

peterson2001@tlen.pl

Independent Researcher, Łódź, Poland

Aleksandra Szymańska, ORCID <https://orcid.org/0009-0002-2122-7522>

ola.szymanska9000@tlen.pl

Independent Researcher, Łódź, Poland

Corresponding author

Aleksandra Dobrzańska, aleksandra.dobrzanska89@gmail.com

Abstract

Background: Dental professionals are at high risk of work-related musculoskeletal disorders (MSDs), especially affecting the neck, lower back, shoulders and upper extremities. Exercise-based interventions may help prevent and manage these conditions, but evidence in dental professionals remains limited.

Aim: To evaluate the efficacy of exercise-based interventions for the prevention and management of musculoskeletal symptoms among dental professionals.

Material and methods: Literature published between 2003 and 2025 was reviewed using international scientific databases, primarily PubMed. Intervention, longitudinal, and cross-sectional studies, as well as systematic reviews and meta-analyses, were considered.

Results: Resistance-based and combined exercise programmes show beneficial effects on musculoskeletal outcomes, including pain, muscular capacity, endurance and disability. Evidence from broader populations with chronic low back pain also supports Pilates, strength, core-based and mind-body exercises, although these outcomes cannot be directly generalised to dental professionals. Optimal exercise dosage and long-term preventive effects remain insufficiently established.

Conclusions: Exercise-based interventions may be an important component of strategies for managing and preventing work-related MSDs in dentistry. Resistance and combined programmes show promising effects on pain and functional outcomes. Further well-designed studies are needed to establish optimal exercise dosage and long-term effectiveness. Integrating exercise and ergonomic education into dental training may support preventive behaviours and long-term musculoskeletal health.

Keywords: dental professionals, musculoskeletal disorders, occupational health, resistance training, exercise therapy

1. Introduction

Dental practice exposes clinicians to substantial biomechanical and psychosocial demands and is associated with a high prevalence of work-related musculoskeletal disorders (MSDs) (Soo et al., 2023; De Sio et al., 2018). This clinical strain is not exclusive to experienced practitioners. Longitudinal observations indicate that musculoskeletal symptoms may emerge during undergraduate dental training, potentially predisposing students to persistent symptoms later in their professional careers (Hodačová et al., 2022; Ohlendorf et al., 2020). Prevalence estimates frequently exceed 70% in the epidemiological literature, underscoring MSDs as a major occupational health concern that may contribute to reduced work capacity and, in some cases, premature retirement (Alzayani et al., 2021; Lietz et al., 2018; Danylak et al., 2024). A major contributor to this morbidity is the physical configuration of the dental operator, which may require practitioners to maintain prolonged static and awkward positions. (De Sio et al., 2018; Rana et al., 2024). The anatomical regions most commonly affected include the cervical and lumbar spine and the shoulder complex (Lietz et al., 2018; Soo et al., 2023).

Despite growing clinical interest, intervention studies remain highly fragmented. Methodological heterogeneity remains a major limitation of the current evidence base, making

it difficult to establish standardised exercise-based prevention protocols (Danylak et al., 2024; Mulimani et al., 2018). Crucially, the optimal exercise dose—including weekly frequency, intensity, volume, and session duration—remains insufficiently defined for the prevention and management of MSDs specifically in dental professionals (Danylak et al., 2024). Long-term studies evaluating adherence to and maintenance of multimodal exercise programmes also remain limited (Danylak et al., 2024; Alkan et al., 2025). Therefore, this review synthesises current evidence on exercise-based MSD prevention among dental professionals and evaluates the relative efficacy of distinct training modalities.

Research Objective. This review evaluates the efficacy of active exercise-based strategies and their specific prescriptive parameters in reducing the clinical burden and functional limitations associated with work-related musculoskeletal disorders in dental practitioners.

Research Problems. Despite the availability of ergonomic interventions, musculoskeletal symptoms remain highly prevalent among dental professionals. Existing literature indicates considerable variation in the implementation and evaluation of ergonomic and exercise-based preventive strategies.

Research Hypotheses. It is hypothesised that active physical conditioning protocols incorporating progressive resistance and core stabilisation will yield greater improvements in musculoskeletal health outcomes than passive ergonomic measures alone (Alkan et al., 2025; Holzgreve et al., 2022; Maestroni et al., 2020).

2. Research materials and methods

This study synthesises peer-reviewed scientific studies, including empirical longitudinal investigations, systematic reviews, and meta-analyses addressing musculoskeletal health, occupational risk factors, and exercise-based interventions relevant to dental professionals (Danylak et al., 2024; Lietz et al., 2018; Roll et al., 2019; Owen et al., 2020). Available literature provides a broad but heterogeneous evidence base concerning occupational musculoskeletal strain and the effects of physical and ergonomic interventions, including resistance training and exercise-based approaches (Alkan et al., 2025; Roll et al., 2019; Holzgreve et al., 2022). A qualitative comparative synthesis was used to examine patterns and differences across clinical trials, longitudinal studies, and observational research, with particular attention to exercise modality, frequency, duration, intensity, adherence, and reported functional outcomes (Danylak et al., 2024; Lietz et al., 2018; Lietz et al., 2020).

Artificial intelligence tools were used for basic data analysis, language improvement, text formatting and to increase efficiency in pattern recognition. All authors reviewed and approved the final version of the manuscript, ensuring its academic integrity and accepting full responsibility for its content.

3. Research results

3.1 Biomechanical Burden & Targeted Anatomical Structures

Precision dental procedures require prolonged static muscle loading within a restricted operative field (De Sio et al., 2018; Lietz et al., 2018), often necessitating sustained awkward postures such as pronounced neck flexion and asymmetric trunk rotation. This state increases mechanical stress of the affected tissues and may contribute to local muscle fatigue and discomfort (Kumar et al., 2014; Surve et al., 2022). Cumulative mechanical exposure from prolonged static and constrained postures may impair muscular performance, leading to persistent pain and long-term functional limitations (Rana et al., 2024; Surve et al., 2022; Kumar et al., 2014; Holzgreve et al., 2022).

Musculoskeletal strain in dental professionals commonly affects anatomical regions involved in maintaining prolonged working postures. The cervical spine is particularly vulnerable. Sustained seated postures may increase mechanical stress of the lumbar region and contribute to low back discomfort, particularly when combined with inadequate postural control (Hodges, 2003; Hodačová et al., 2022). The shoulder region is also frequently affected, with prolonged elevation and abduction contributing to increased ergonomic risk and upper-extremity musculoskeletal symptoms (Kumar et al., 2014; Rafie et al., 2015).

While magnification loupes, prismatic eyewear, and adjustable operator seating can improve working posture and reduce some ergonomic exposures, these interventions do not directly provide the muscular adaptations associated with exercise training (Danylak et al., 2024; Lietz et al., 2020). Consequently, active exercise interventions represent a potentially important complementary strategy for improving physical capacity, muscular endurance, and neuromuscular control (Alkan et al., 2025; Maestroni et al., 2020).

General physical activity may not fully address the specific muscular endurance and motor-control requirements associated with prolonged dental postures. Aerobic conditioning primarily supports general physical fitness, whereas targeted strengthening and mobility exercises may address the specific muscular demands associated with prolonged dental postures (Kumar et al., 2014; Roll et al., 2019). The principle of training specificity suggests that

adaptations are influenced by the muscles, movement patterns, and physiological demands specifically targeted during training. Consequently, targeted exercise may be particularly relevant when the aim is to improve specific aspects of cervical and trunk muscular endurance (Alkan et al., 2025; Kawtharani et al., 2023). Targeted resistance training enhances maximal strength (MVIC) and muscular capacity, effectively lowering the relative physiological strain during clinical tasks (Holzgreve et al., 2022; Maestroni et al., 2020). A combined approach incorporating mobility exercises, strengthening, and motor-control training may be appropriate for addressing the flexibility, strength, and endurance demands. (Hodges, 2003; Kumar et al., 2014). Improving trunk and cervical strength may enhance tolerance to the physical workload of dental practice and could contribute to long-term musculoskeletal health (Holzgreve et al., 2022; Kumar et al., 2014).

3.2 Resistance & Strength Training Protocols

Clinical trials indicate that exercise interventions focusing on the trunk, cervical, and scapular musculature significantly increases muscular strength and endurance, thereby improving postural capacity during clinical procedures (Alkan et al., 2025; Holzgreve et al., 2022; Kumar et al., 2014). Those exercise programmes should use appropriately scaled resistance and progressive loading to promote adaptation while allowing adequate recovery (Maestroni et al., 2020; Phillips and Winett, 2010).

In adults with chronic low back pain, mind-body and core-based physical therapy interventions, including Pilates, can improve pain and functional disability (Fernández-Rodríguez et al., 2022; Owen et al., 2020). These findings may be relevant when considering exercise strategies for dental professionals, although direct evidence in dentists remains limited. Core-based exercise protocols may target trunk motor control and the activation and endurance of deep trunk musculature, including the multifidus and transversus abdominis (Hodges, 2003; Owen et al., 2020). Yoga-based physical activity has also been proposed as a strategy for addressing musculoskeletal symptoms among dental professionals (Koneru and Tanikonda, 2015; Roll et al., 2019). Together, these approaches may provide useful components of exercise-based rehabilitation and musculoskeletal symptom management, although their specific preventive effectiveness in dental professionals requires further investigation (Hodges, 2003; Owen et al., 2020).

3.3 Intervention Parameters, Dosage, and Compliance

Achieving meaningful muscular adaptation requires meeting specific exercise thresholds. Clinical studies in dental professionals have demonstrated beneficial effects of exercise programmes lasting 10 to 12 weeks. A 10-week resistance-training programme improved pain and MVIC, while a 12-week combined resistance, stretching and postural-alignment programme improved pain, muscular endurance and disability outcomes (Holzgreve et al., 2022; Alkan et al., 2025). Exercise regimes evaluated in dental professionals have typically used frequencies of two to three sessions per week (Alkan et al., 2025; Holzgreve et al., 2022). In adults with chronic low back pain, a network meta-analysis identified programmes involving at least one to two weekly sessions of Pilates or strength exercise among the most beneficial intervention patterns (Fernández-Rodríguez et al., 2022). However, the applicability of these conclusions to dental professionals remains uncertain. Intervention duration has varied across studies. Holzgreve et al. (2022) used 60-minute resistance-training sessions, whereas the telerehabilitation programme evaluated by Alkan et al. (2025) used 45-minute sessions. These results indicate that exercise programmes of different session durations may be viable, although direct comparisons of their effectiveness are lacking. Aerobic exercise may complement resistance and flexibility training by supporting general physical fitness and providing a whole-body component of an exercise-based prevention programme (Kumar et al., 2014). Stretching may be incorporated into exercise-based prevention programmes to address muscular tightness. In the study by Alkan et al. (2025), stretching exercises were performed for 20 seconds per repetition as part of a combined 12-week programme.

Defining appropriate exercise parameters is important for translating exercise-based interventions into reproducible clinical protocols; however, the optimal combination of frequency, intensity, duration and exercise volume for dental professionals remains insufficiently established (Alkan et al., 2025; Holzgreve et al., 2022; Maestroni et al., 2020). Current evidence suggests that appropriately dosed resistance and combined exercise programmes can improve muscular capacity and reduce pain or disability outcomes; however, the optimal training volume and intensity for dental professionals remain uncertain (Alkan et al., 2025; Holzgreve et al., 2022). As detailed in Table 1, the synthesized exercise categories, anatomical targets and intervention parameters reported in the literature provide a framework for developing exercise-based approaches to the prevention and management of work-related musculoskeletal symptoms among dental professionals (Alkan et al., 2025; Maestroni et al., 2020; Roll et al., 2019).

Table 1. Summary of Evidence-Based Exercise Modalities, Biomechanical Objectives, and Prescriptive Parameters for Dental Professionals

Exercise Category	Targeted Anatomical Regions & Muscles	Key Biomechanical Objectives	Recommended Parameters/Dosage	Supporting Sources
Progressive Resistance Training	Cervical stabilizers, scapular retractors (middle/lower trapezius), serratus anterior, and trunk stabilizers.	Augmenting maximal strength potential (MVIC); enhancing postural endurance; reducing relative physiological load.	2–3 sessions per week; 20–60 minutes; 3 sets of 10–12 repetitions using elastic bands or weights.	Alkan et al., 2025; Holzgreve et al., 2022; Maestroni et al., 2020; Phillips and Winett, 2010
Static & Dynamic Stretching	Upper trapezius, levator scapulae, pectoralis major, hamstrings, and hip flexors.	Lengthening pathologically shortened structures; alleviating hypertonicity; promoting nutrient blood flow to	Slow, controlled holds of 15–30 seconds; executed 2–3 times daily or during intra-operative intervals.	De Sio et al., 2018; Kumar et al., 2014; Rafeemanesh et al., 2025

		ischemic tissues.		
Intra-Workday Micro-Breaks	Global spinal structures and upper extremity kinetic chains.	Interrupting cumulative static loading; facilitating metabolic waste removal; immediate reduction of muscular fatigue.	30–50 second intervals between patient sessions; incorporating isometric pauses or chair-side stretches.	Kumar et al., 2014; Lietz et al., 2020; Rafeemanesh et al., 2025
Mind-Body Modalities (Pilates/Yoga)	Deep spinal stabilizers (multifidus, transversus abdominis) and core musculature.	Optimizing neuromuscular motor control; enhancing proprioception ; restoring natural spinal alignment and lordosis.	1–2 sessions per week; sessions <60 minutes; minimum program duration of 3–9 weeks.	Álvarez-Bueno et al., 2022; Hodges, 2003; Koneru and Tanikonda, 2015; Owen et al., 2020

Aerobic Conditioning	Systemic musculoskeletal and cardiopulmonary systems.	Improving oxygen transport and efficiency; facilitating nutrient delivery to overtaxed muscular tissues.	3–4 times per week; 20–30 minutes per session; intensity at ~70% of maximal heart rate.	Kumar et al., 2014; Maestroni et al., 2020; Roll et al., 2019
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Despite the potential health benefits of regular physical activity, dental practitioners may face personal and workplace barriers that limit regular exercise. High patient loads, extended working hours and limited opportunities for breaks may make it difficult for clinicians to incorporate regular exercise into the working day (Kumar et al., 2014; Pejčić et al., 2017). Psychological stress and insufficient physical activity have both been associated with a higher prevalence of musculoskeletal disorders among dentists (Surve et al., 2022). Limited ergonomic and preventive education may represent an additional barrier to effective MSD prevention, particularly during the transition from dental education to professional practice (Hodačová et al., 2022; Danylak et al., 2024). Ergonomically inadequate work environments and inconsistent implementation of preventive measures may further limit effective MSD prevention in dental practice (De Sio et al., 2018; Danylak et al., 2024).

Simple, flexible exercise programmes that can be incorporated within daily routines may improve the feasibility of long-term adherence (Kumar et al., 2014; Phillips and Winnett, 2010). Telerehabilitation may provide an accessible alternative to fully supervised exercise protocols, particularly for clinicians with limited access to face-to-face rehabilitation (Alkan et al., 2025). Both supervised and telerehabilitation-based exercise programmes produced beneficial outcomes in dentists with chronic neck pain, suggesting that telerehabilitation may be an attainable alternative when face-to-face supervision is difficult to access (Alkan et al., 2025). Integrating ergonomic education and preventive exercise strategies into undergraduate

dental training may provide an opportunity to establish preventive behaviours before students enter professional practice (Hodačová et al., 2022; Danylak et al., 2024).

4. Discussion

Effective prevention of work-related musculoskeletal disorders in dentistry is likely to require a combination of ergonomic, behavioural and physical strategies rather than reliance on a single intervention (De Sio et al., 2018; Lietz et al., 2020). Evidence from separate intervention studies supports both ergonomic modifications and structured exercise as potentially beneficial approaches, although the additional benefit of combining these strategies has not been adequately established (Lietz et al., 2020; Roll et al., 2019; Alkan et al., 2025). A multimodal approach may address several modifiable contributors to occupational musculoskeletal strain, including working posture, ergonomic exposure and physical capacity (Holzgreve et al., 2022; Kumar et al., 2014). It combines modification of external working conditions with improvements in physical capacity that may help clinicians tolerate occupational postural demands (Holzgreve et al., 2022; Kumar et al., 2014). Taken together, the available evidence supports incorporating regular physical conditioning and postural awareness alongside ergonomic modifications rather than relying exclusively on passive equipment (Alkan et al., 2025; Holzgreve et al., 2022).

Despite a growing body of literature, methodological weaknesses continue to limit the strength and generalizability of the evidence. Marked heterogeneity across published studies makes it difficult to establish universal clinical standards (Danylak et al., 2024; Lietz et al., 2020). A substantial proportion of the evidence derives from cross-sectional studies using self-reported questionnaires, which limits causal inference and introduces the potential for recall and reporting bias, thereby compromising the reliability of prevalence estimates and symptom assessment (De Sio et al., 2018; Rafeemanesh et al., 2025). Furthermore, the small sample sizes and short follow-up periods of many intervention studies limit the strength of the available evidence. Long-term maintenance of exercise-related benefits remains insufficiently established, as recent dental exercise trials have primarily evaluated outcomes over approximately 10–12 weeks (Alkan et al., 2025; Holzgreve et al., 2022). The optimal combination of exercise intensity, session duration, and weekly frequency for dental professionals also remains uncertain, and evidence from broader chronic low back pain populations cannot be assumed to apply directly to dental practice (Fernández-Rodríguez et al., 2022; Alkan et al., 2025; Holzgreve et al., 2022). Few controlled studies have examined combined physical, psychosocial, and organizational interventions, leaving the interaction

between exercise, stress management, and workflow modification poorly understood (Danylak et al., 2024; Roll et al., 2019).

Subsequent research should complement subjective survey data with objective physical measurements (Danylak et al., 2024; Ohlendorf et al., 2020). Wearable motion sensors, surface electromyography (EMG) and other biofeedback technologies may enable more objective assessment of posture, muscle activity and physical loading during clinical work (Danylak et al., 2024; Ohlendorf et al., 2020). Large-scale prospective cohort studies are needed to determine whether preventive ergonomic and exercise education introduced during dental training produces benefits that persist into professional practice (Hodačová et al., 2022; Danylak et al., 2024). Future trials should establish exercise dosage parameters specifically for dental professionals and evaluate whether digital health and telerehabilitation approaches can sustain adherence and clinical benefits over longer follow-up periods (Alkan et al., 2025; Danylak et al., 2024). Integrating motion-tracking technologies into clinical simulation laboratories could provide an objective platform for assessing posture and ergonomic risk and may strengthen the evidence base for primary MSD prevention (Danylak et al., 2024; Ohlendorf et al., 2020).

5. Conclusions

Work-related musculoskeletal disorders are highly prevalent in dentistry but may be partially preventable through modifiable ergonomic and physical interventions. Emerging evidence suggests that structured exercise combining strengthening, stretching, and postural-control exercises can help reduce discomfort and improve physical function in dental professionals.

Mobility and stretching exercises may help maintain range of motion and reduce muscular tightness associated with sustained clinical postures. Together, these interventions may support musculoskeletal health and improve physical capacity, although their ability to prevent the progression from occupational fatigue to chronic injury remains insufficiently established.

Dental schools should consider incorporating structured exercise and posture education into undergraduate training, particularly because musculoskeletal symptoms may develop during dental education. Early instruction may help students develop preventive self-care behaviours before musculoskeletal symptoms become established.

During daily practice, clinicians may use brief chairside stretching, mobility, or postural exercises to interrupt prolonged static loading, reduce perceived muscular discomfort, and

restore a more neutral working posture. Digital exercise-delivery tools, including telerehabilitation, may also facilitate regular participation among clinicians facing demanding work schedules. A shift from predominantly reactive management toward proactive, long-term prevention that combines ergonomic modifications with regular physical conditioning could improve the musculoskeletal health and career longevity of dental professionals.

Disclosure

Supplementary Materials

Not applicable

Author Contributions

All authors have read and agreed to the published version of the manuscript.

Conceptualization: Aleksandra Dobrzańska, Klaudia Borycka, Aleksandra Baranowska

Methodology: Maria Berek, Anna Hajduk, Piotr Szymański

Investigation: Kacper Kmieciak, Jagoda Stańska, Klaudia Borycka

Formal analysis: Filip Kostanek, Jagoda Stańska, Aleksandra Baranowska

Writing - original draft preparation: Aleksandra Dobrzańska, Anna Hajduk, Filip Kostanek

Data curation: Aleksandra Szymańska, Filip Kostanek, Anna Hajduk

Visualization: Maria Berek, Kacper Kmieciak, Aleksandra Dobrzańska

Project administration: Aleksandra Szymańska, Piotr Szymański

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Conflicts of Interest

The authors deny any conflict of interest.

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